American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science &Technology (FST)

Spring: 2025 - 2026

Section: A

Software Quality Assurance and Testing

MediCare

Reported Submitted By:

SN	Name	ID
1	MD. Mridul Ali	22-48929-3

Checked By Industry Personnel:

Name:

Designation:

Company:

Sign:

Date:

Date:

Software Test Plan

For

MediCare

Version 2.0

Mridul Ali .MD :Prepared by

American International University-Bangladesh

07/05/2026 :Date Created

1. TEST PLAN IDENTIFIER: MediCare-VER00.2

2. REFERENCES

I. Software Requirement Specification (SRS) of MediCare
II. Course Material: Software Quality Assurance and Testing

3. PROBLEM DOMAIN

3.1 Background to the Problem:

Hospital Access and Documentation: In hospitals, patients are often required to carry physical documents detailing their medical history and current/future treatment plans. This can be burdensome for patients, especially in urgent or emergency situations where immediate access to such documents is difficult.

Limited Treatment Options: In critical circumstances, individuals may not be able to find a suitable hospital or doctor to administer the necessary treatment. This highlights a significant gap in the availability of efficient medical services and timely treatment, which is seen as a major problem.

Healthcare in Rural Areas: Many rural areas in our country suffer from significant health related issues but lack access to proper treatment. The remote nature of these areas and the inability of medical professionals to reach or properly diagnose the patients exacerbate this issue.

Challenges for Senior Citizens: Senior citizens face difficulties in visiting hospitals, especially for quick treatments. Due to age and health constraints, they may struggle to travel or receive timely care. This societal issue requires urgent attention.

3.2 Solution to the Problem:

Objective: The main objective of this project is to create a technological solution that streamlines access to medical history, simplifies healthcare access for rural populations, and provides an efficient method for senior citizens to receive care without constant hospital visits.

Proposed Solutions:

Digital Medical Records: This solution enables patients to store and access their medical history and treatment plans digitally, eliminating the burden of carrying physical documents and ensuring healthcare providers have immediate access to the necessary information for timely, accurate treatment.

Telemedicine: By integrating telemedicine features, patients in remote areas can consult with doctors through video calls or other virtual means, helping bridge the gap in healthcare access.

Mobile Healthcare Services for Seniors: For senior citizens, the solution could include mobile healthcare services that allow medical professionals to visit patients at home or use remote monitoring technologies to track health metrics and alert healthcare providers when intervention is needed.

Emergency Ambulance Service: An emergency ambulance service feature will be integrated into the platform, ensuring that patients who require urgent medical attention can quickly request an ambulance. This feature would include real-time tracking, faster dispatch times, and integration with local emergency services to ensure a swift response, particularly for remote or underserved areas.

Feasibility: The proposed solutions are highly feasible, leveraging current technological advancements such as cloud storage, telemedicine, mobile health apps, and emergency service integration, which are already being implemented in various parts of the world.

Existing Software Solutions: Existing solutions for digital health records (e.g., Epic) and telemedicine (e.g., Teladoc) typically operate separately. This project aims to integrate these functionalities into one comprehensive platform, while also including emergency ambulance services for a better healthcare solution.

Extension of Existing Solutions: While current solutions provide some of the necessary functionalities, this project extends them by offering a more user-friendly, comprehensive system on a single platform that targets the unique needs of rural and elderly populations and includes emergency services for critical care.

4. REQUIREMENT SPECIFICATION

4.1 System Features

1. **Emergency Login (Quick Access)**
   1.1 Users must be able to log in using only their phone number and OTP during an emergency.
   1.2 Emergency login must allow immediate access to request ambulance and hospital services for free.
   1.3 No password or additional data should be required during emergency login.
   1.4 Emergency login access must be time-limited or session-limited for security.

2. **Regular Sign-Up and Login**
   2.1 Users must be able to sign up with a phone number and password during regular (non-emergency) times.
   2.2 After first-time login, users must fill in additional personal information (e.g., name, Address, age, medical history).
   2.3 Login credentials must be securely encrypted.
   2.4 Users must have role-based access (patient, doctor, ambulance staff).

3. **Emergency Ambulance and Hospital Service**
   3.1 Emergency ambulance requests must be available immediately after emergency login.
   3.2 Ambulance location must be tracked in real-time on the platform.
   3.3 Emergency hospital service (nearby hospital finder) must be provided instantly after emergency login.
   3.4 Emergency services must be free of cost during emergency login.

4. **Digital Medical Records/Documentation (Paid Feature)**
   4.1 Users must have the option to purchase access to digital medical records.
   4.2 After purchasing, users can store and manage their complete health history digitally.
   4.3 Doctors must be able to view and update these records securely.
   4.4 Users must be able to download their documentation in PDF or other standard formats, but this will only be available for premium users.
   4.5 The documentation feature should be easy to navigate and present medical records in a clear, readable format.

5. **Telemedicine (Remote Consultation)**
   5.1 Patients must be able to schedule and attend video consultations with doctors.
   5.2 Doctors must access patient history during the consultation.
   5.3 Video and chat communications must be encrypted.

6. **Language Options**
   6.1 Users must be able to select their preferred language (e.g., English, Bengali).
   6.2 Language preference must be saved in the user's profile.
   6.3 Default language must be selectable at the time of first login or emergency access.

7. **Notifications and Alerts**
   7.1 Appointment reminders must be sent to patients and doctors.
   7.2 Emergency alerts must notify nearby ambulances and hospitals.
   7.3 Medicine reminders must be sent to subscribed users.

8. **Mobile Healthcare for Senior Citizens**
   8.1 Senior citizens must be able to request home-based doctor visits.
   8.2 Healthcare staff must receive real-time alerts for home visit requests.
   8.3 Remote monitoring devices must sync health data automatically.

9. **Search**
   9.1 Users must be able to search for nearby hospitals and doctors.
   9.2 Search results must display availability status and estimated distance.

4.2 System Quality Attributes

1. **Security**

    1.1 All user data must be protected by encryption.

    1.2 Login sessions must be safe and use secure connections (SSL/TLS).

    1.3 Emergency login should automatically log out after a short time to avoid misuse.

    1.4 The system must follow healthcare data protection rules (like HIPAA or GDPR).

2. **Performance**

    2.1 Emergency services must be loaded within 2 seconds.

    2.2 The app must handle at least 1,000 users at the same time without crashing.

    2.3 Ambulance location updates must be refreshed every 5 seconds.

    2.4 Medical records must open within 3 seconds after clicking.

3. **Usability**

    3.1 The app must be simple and easy for everyone to use, even in emergencies.

    3.2 Users must be able to choose their language easily.

    3.3 Big text, big buttons, and clear colors must be used for elderly people.

    3.4 Emergency actions (like calling an ambulance) must be possible in just 2 clicks.

4. **System Performance and Availability**

    4.1 System must respond to user actions within 2 seconds.

    4.2 Platform availability must be at least 99.9% uptime.

5. **Reliability**

    5.1 Automatic backups must happen every day.

    5.2 If the server crashes, it must recover in less than 1 minute.

6. **Maintainability**

    6.1 The app must allow easy updates without downtime.

    6.2 The code must be clean and well-documented for future developers.

    6.3 Major bugs must be fixed and updated within 24 hours.

7. **Scalability**

    7.1 The app must support at least 10,000 users easily.

    7.2 During big emergencies, the system must automatically add more servers if needed.

4.3 System Interface

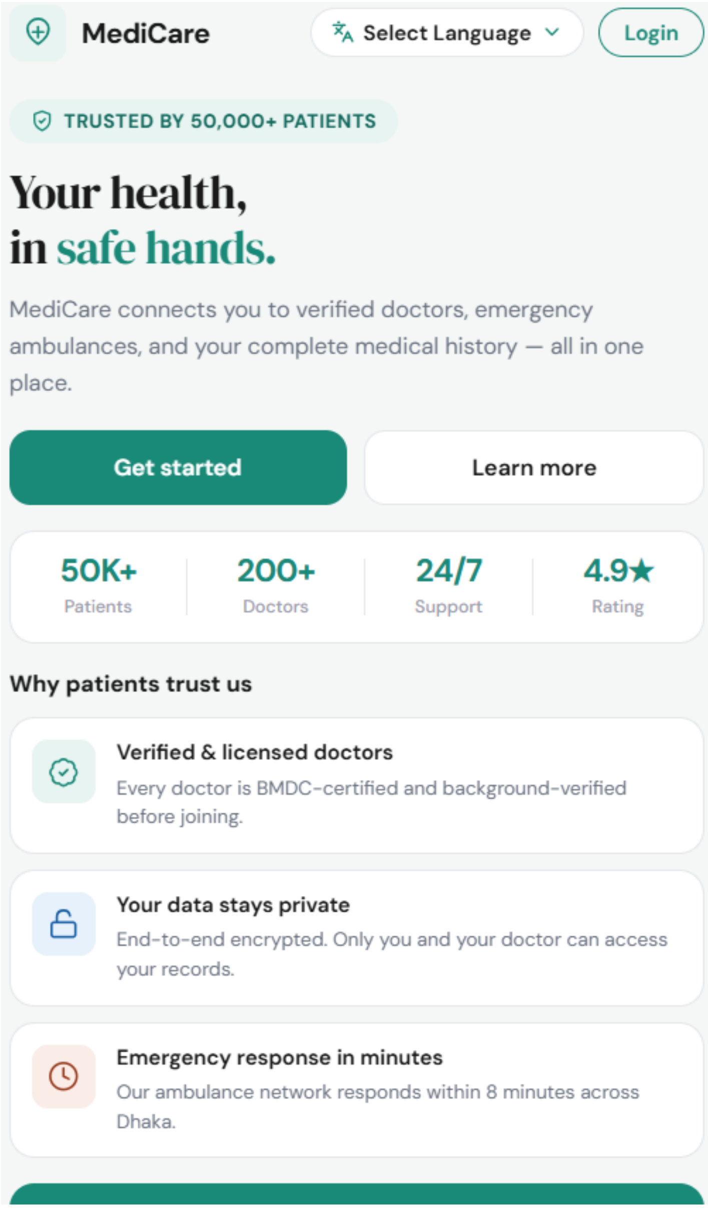


Figure 1: Landing Page

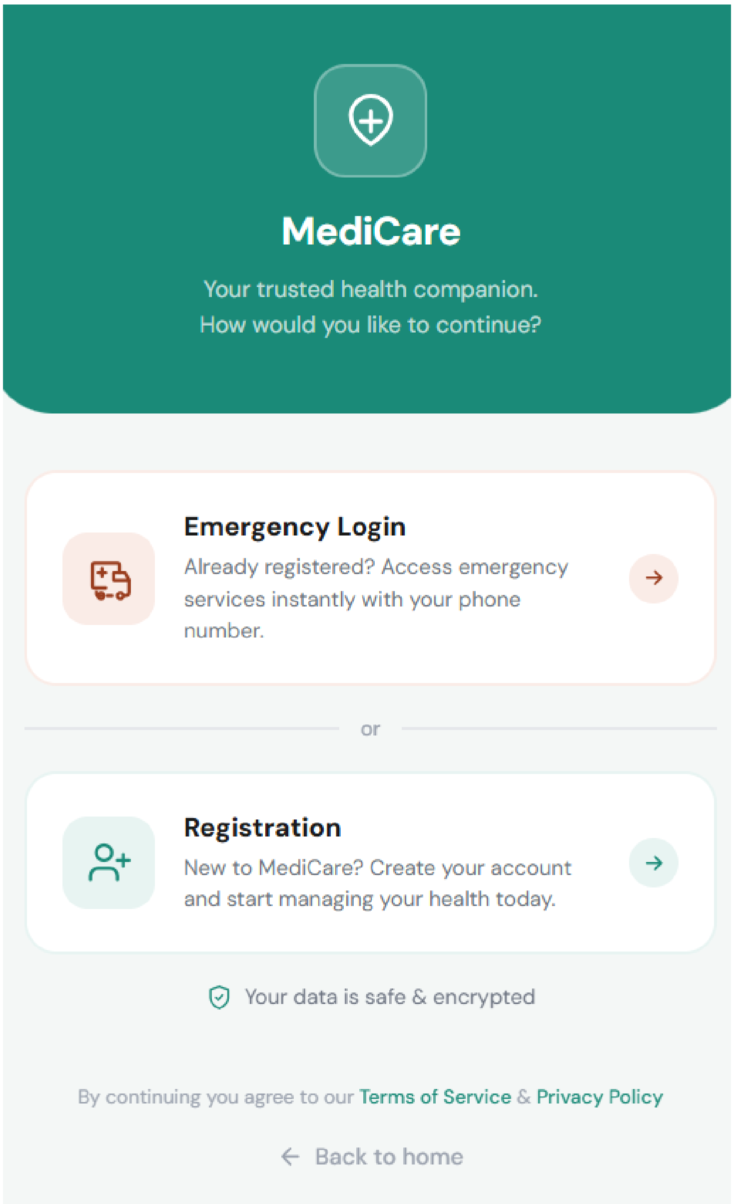


Figure 2: Login Options

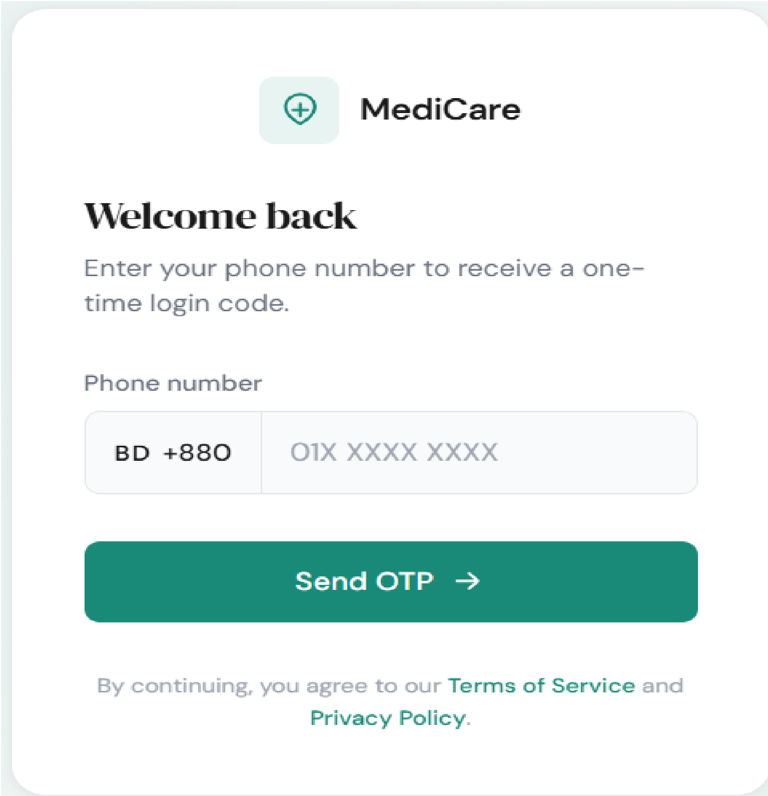


Figure 3: Emergency Login Page

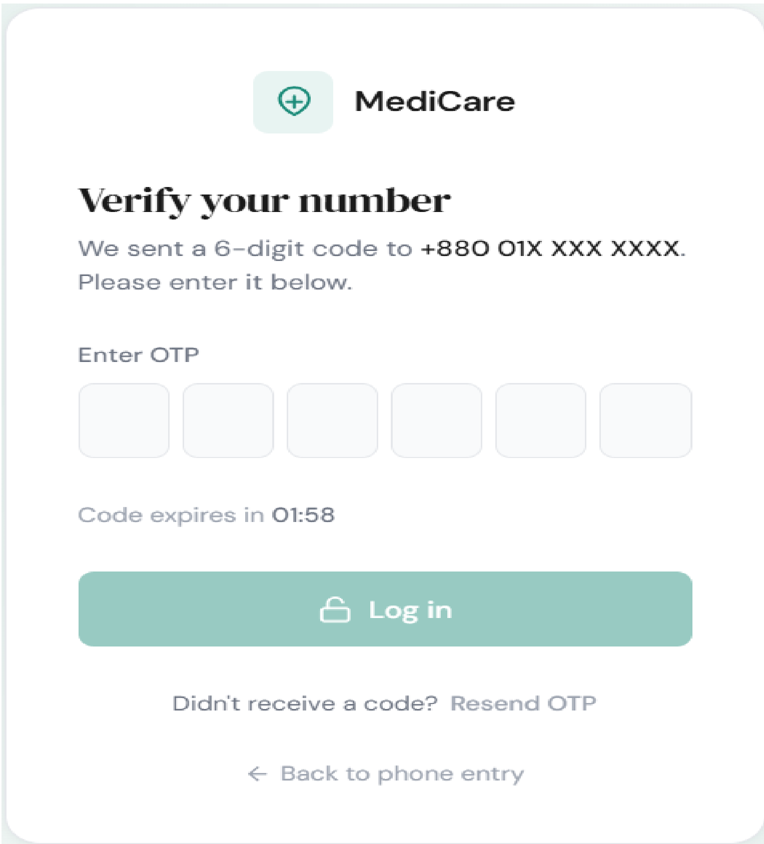


Figure 4: Verify Phone Number

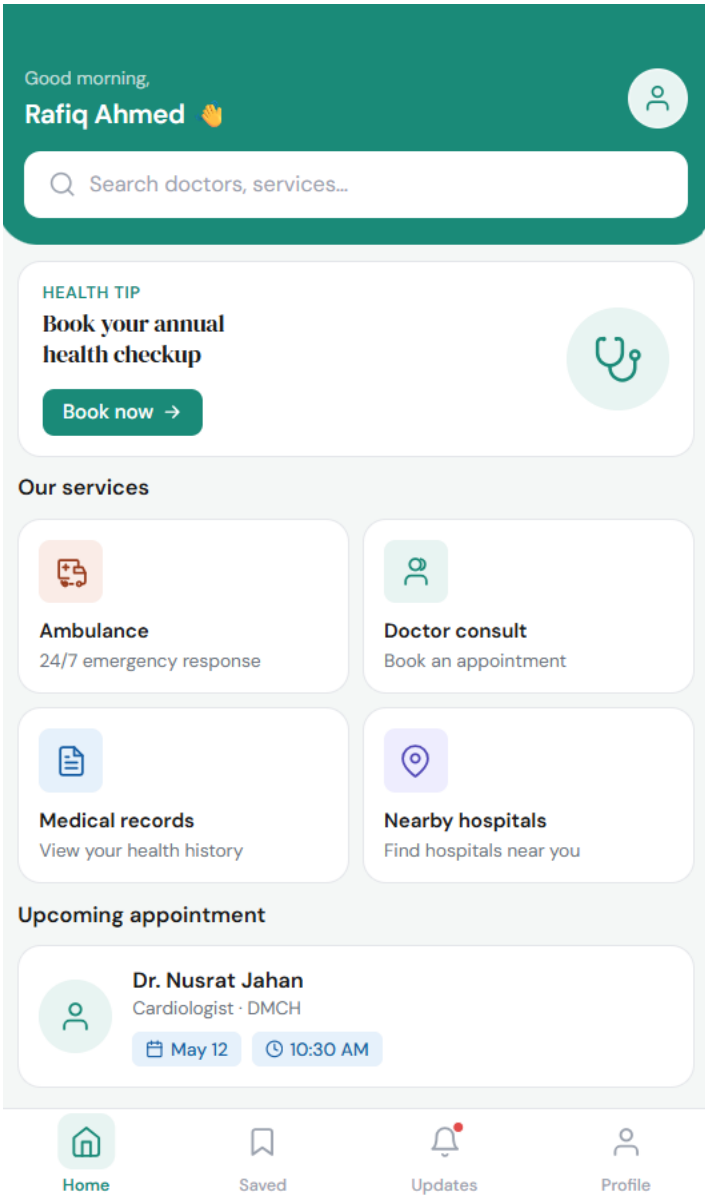


Figure 5: Home Page

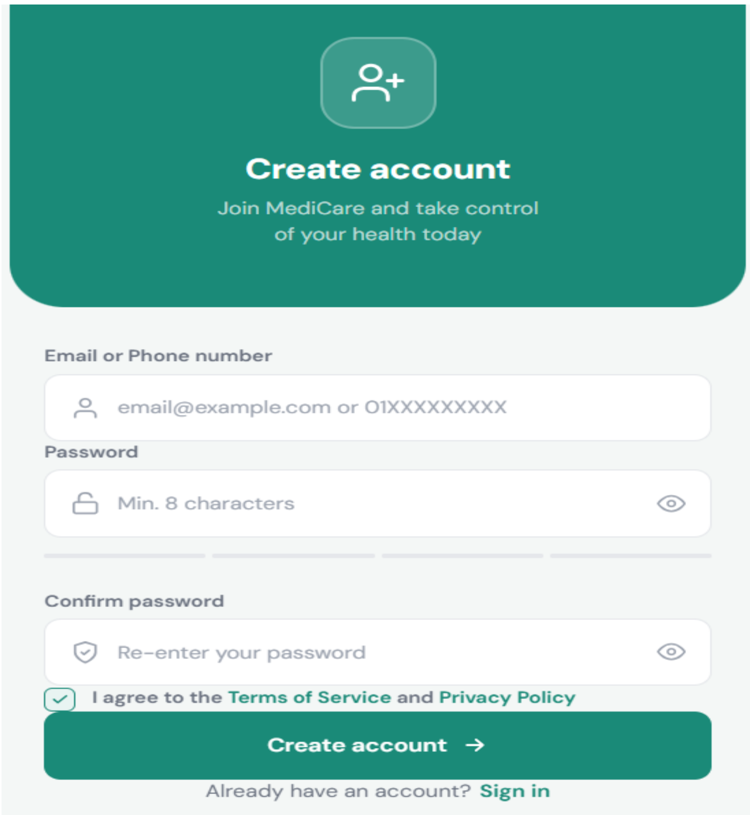


Figure 6: Registration Page

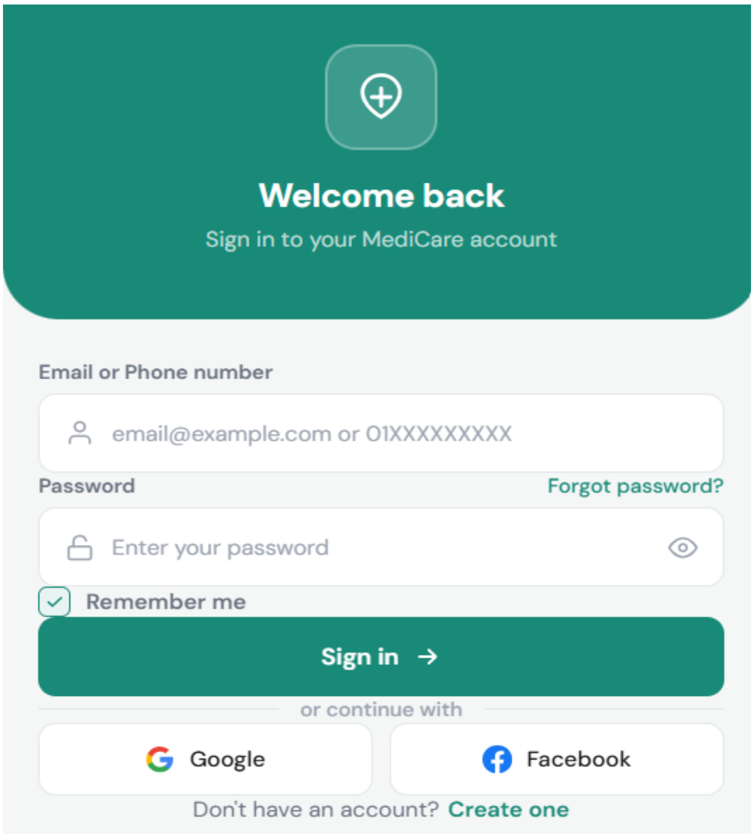


Figure 7: Login Page

4.4 Project Requirements

4.4.1 Effort Estimation

Project category is Semi-Detached. Because medium size, moderately complex and team have mixed experience.

Software Project Type	Coefficient <Effort Factor>	P	T
Organic	2.4	1.05	0.38
Semi-detached	3.0	1.12	0.35
Embedded	3.6	1.20	0.32

PM: person-months needed for project (labor working hours)

SLOC: source lines of code = 50,000 (As this is a medium size project)

P: project complexity = 1.12 (semidetached)

DM: duration time in months for project (weekdays)

T: SLOC-dependent coefficient =0.35(semidetached)

ST: average staffing necessary.

Effort = PM = Coefficient $_{\text{<Effect Factor>}}$*(SLOC/1000) ^ P

= 3 * (50000/1000) ^ 1.12

= 239.86 Person-Months (labor working hours)

Development Time = DM = 2.50*(PM) ^ T

= 2.50*(239.86) ^ 0.35

= 17.02 Months (Weekdays)

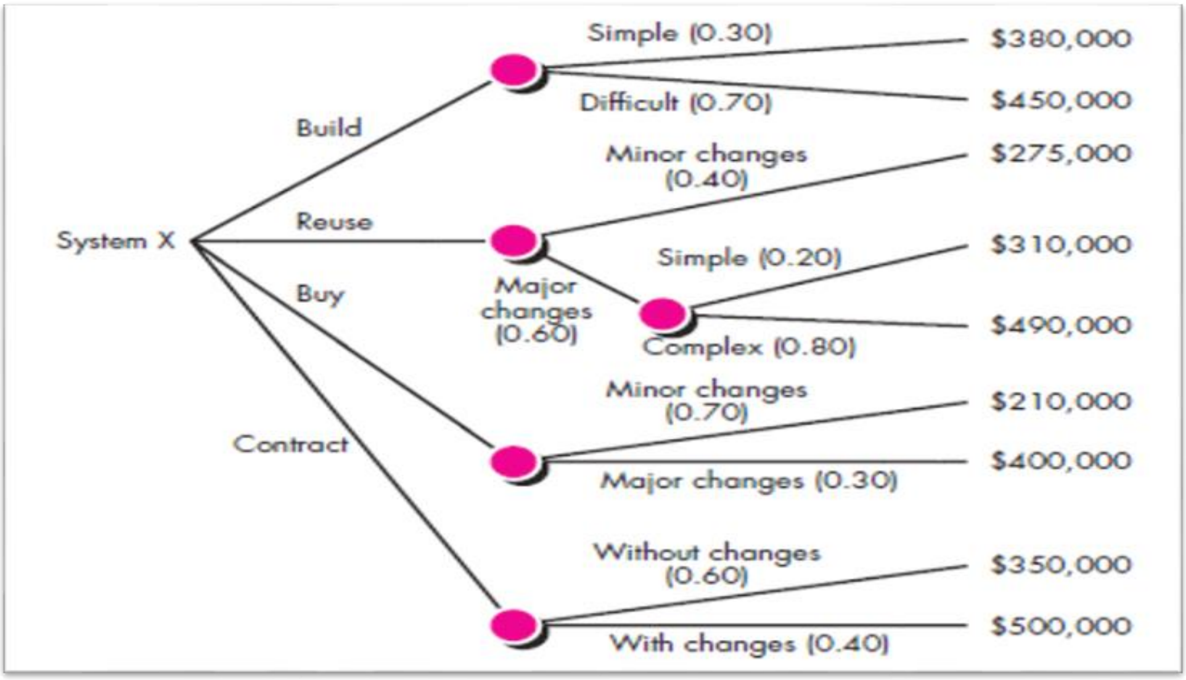
Required number of people = ST = PM/DM

= 239.86 / 17.02

= 14.1 ≈ 14

4.4.2 Budget Estimation

Computing expected cost from DECISION TREE



expected cost = $(\text{path probability})_i$ * $(\text{estimated path cost})_i$

expected cost $_{build}$ = 0.3 * $380k = $114k

expected cost $_{reuse}$ = 0.4 * $275k = $110k

expected cost $_{buy}$ = 0.7 * $210k = $147k

expected cost $_{contra}$ = 0.4 * 500k = $200k

Total cost = $114k + $110k + $147k + $200k = $571k

4.4.3 Resources & Environment

4.4.3.1 Resources

1. **Human Resources**

The following team members and roles are required for the project:

Role	**Responsibility**
Project Manager	Manage project planning and progress
UI/UX Designer	Design user-friendly interface
Frontend Developer	Develop user interface and mobile/web screens
Backend Developer	Develop APIs, authentication, database handling
Database Administrator	Manage medical records and database security
QA/Test Engineer	Perform software testing and bug tracking
DevOps/Cloud Engineer	Handle deployment and server maintenance

Table 1: Human Resources

2. **Software Resources**

Software/Tool	**Purpose**
Visual Studio Code	Code editor
React / Next js	Frontend Development
Node.js / Django / Laravel	Backend Development
MySQL / PostgreSQL	Database Management
Git & GitHub	Version Control
Firebase / Twilio	Notifications and OTP
Zoom/WebRTC API	Video consultation

Table 2: Software Resources

4.4.3.2 Environment

1. Web browser-based System
2. Cloud deployment environment

The System will be deployed using:

- Cloud hosting platform
- Secure HTTPS/SSL connection
- Real-time database synchronization
- Backup and recovery server

3. Network Environment
    - High-speed internet connection
    - Secure API communication
    - GPS and map integration
    - Real-time notification services

5. FEATURES NOT TO BE TESTED

1. Third party Payment gateway internal logic (Bkash, Nagad, Rocket, Visa Card, Master Card etc.)
2. Ambulance service internal system
3. User owned device and browser specific issues
4. External Email/SMS provider infrastructure

6. TESTING APPROACH

6.1 Testing Levels

- **Unit Testing:** Tests a small software unit at a time (Registration, Login, Payment etc.). It is performed by developers and follows the white box testing method.
- **System Integration Testing:** This method is putting modules together in an incremental manner after conducting unit testing on module. The specific QA team will be responsible for this testing approach.
- **System Testing:** It is the testing of a complete and fully integrated software product. It should test the functional and non-functional requirements of the software and specific QA team will be responsible for this testing approach.
- **Load Testing:** The QA team will evaluate system performance under expected peak-users load.
- **Security Testing:** The QA team will verify data protection, access control, and vulnerability handle.
- **Acceptance Testing:** This testing approach verifies functionality and usability of the software, and it is performed by Customers/ End users. It is confirmed that the system meets the agreed upon criteria.

6.2 Test Tools

The MediCare project will use several testing tools to ensure software quality and reliability. Playwright will be used for functional and automated testing of user interfaces. Postman will test backend APIs and emergency service communication. Apache JMeter will perform load and performance testing for concurrent users. OWASP ZAP will be used for security vulnerability testing, and Jira will manage bug tracking and issue reporting.

Testing Type	**Tool**	**Purpose**
Functional Testing	Playwright	Test login, forms, search, emergency requests
API Testing	Postman	Test backend APIs and ambulance services
Performance/Load Testing	Apache JMeter	Test 1000+ concurrent users
Security Testing	OWASP ZAP	Detect security vulnerabilities
Database Testing	MySQL Workbench	Verify database records
Bug Tracking	Jira	Track bugs and project tasks

Table 3: Testing Tools

6.3 Meetings

Regular meetings will be conducted throughout the MediCare project lifecycle to ensure proper communication, project monitoring, risk management, and software quality assurance. Different types of meetings will be arranged based on project needs.

Meeting Type	**Participants**	**Frequency**	**Purpose**
Project Initial Meeting	Project Manager, Developers, QA Team, Stakeholders	Once at project start	Discuss project goals, scope, requirements, timeline, and responsibilities
Requirement Review Meeting	Project Manager, Developers, QA Team, Client/Stakeholders	Weekly (during requirement phase)	Review and confirm functional and non-functional requirements
Development Progress Meeting	Project Manager, Frontend & Backend Developers	Weekly	Monitor development progress, task completion, and technical issues

QA/Test Planning Meeting	QA Team, Developers, Project Manager	Before testing phase	Finalize test strategy, test cases, tools, and testing schedule
Bug Review Meeting	QA Team and Developers	Twice per week	Discuss detected bugs, severity, priority, and fix status
Sprint/Iteration Meeting	Entire Development Team	Every 2 weeks	Review completed work and plan next iteration tasks
Security & Risk Review Meeting	QA Team, Security Tester, DevOps Engineer	Monthly	Review security vulnerabilities, risks, and mitigation plans
Client Demonstration Meeting	Stakeholders, Project Manager, Development Team	At milestone completion	Demonstrate project features and gather feedback
Final Acceptance Meeting	Client/End Users, QA Team, Project Manager	Before deployment	Confirm system meets requirements and approve final release

Table 4: Meetings

7. TEST CASES/TEST ITEMS

Test Case 1: Emergency Login

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_1</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Emergency Login</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: Verify login with Phone number</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test website Emergency login page</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must have valid Phone Number</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Go to the website
2. Click Login Button
3. Click Emergency Login Button
4. Enter Phone Number
5. Click Send OTP Button</td><td>Phone Number: 013******71</td><td>User should go to the Verify Phone Number Page</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Phone Number is validating User will get the OTP</td></tr>
</table>

Test Case 2: Verify Phone Number

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_2</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Verify Phone Number</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: Verify Phone Number with OTP</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test website Verify Phone Number page</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must have valid Phone Number</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Go to the website
2. Enter OTP
3. Click Login button</td><td>OTP: 589741</td><td>User should Login to the Application</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.</td></tr>
</table>

Test Case 3: Registration

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_3</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Registration Session</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: User can registration with they are Phone Number or Email and Password</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test website Registration Page</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): If User have no Account</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Go to the website
2. Click Login Button
3. Click Registration Button
4. Enter Phone Number
5. Enter Password
6. Click Create Account Button</td><td>Phone Number: 013******71

Email: user@gmail.com

Password: User@103240</td><td>User should go to the Login page</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: User’ Phone Number/Email and Password will store into the Database and successfully create a new Account.</td></tr>
</table>

Test Case 4: Verify Login

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_4			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Login Session			Test Execution date:	
Test Title: Verify Login with valid Phone number/Email and Password				
Description: Test website Registration Page				
Precondition (If any): If User have no Account				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Enter Phone number/Email 3. Enter Password 4. Click Sign In Button	Phone Number: 013******71 Email: user@gmail.com Password: User@417	User should login into the application	As expected,	
Post Condition: User is validated with database and successfully login to account. The account session details are logged into the database.				

Test Case 5: Forgot Password or Change Password

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_5			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Forgot/Chane Password			Test Execution date:	
Test Title: Forgot/Change password with Current Password				
Description: Test website Forgot/Change Password Page				
Precondition (If any): User must have Current password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to profile and click Forgot/ change password 2. Enter Current Password 3. Enter New Password 4. Enter Confirm Password 5. Click Reset Password Button	Current Password: User@417 New Password: Jhon$369 Confirm Password: Jhon$369	After clicking the reset password button user will see the pop-up message (password updated) with Ok Button and after clicking the Ok button user will go to the Home page	As expected,	
Post Condition: New Password will store into the Database				

Test Case 6: Registration with Duplicate Phone Number

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_6	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Registration validation	Test Execution date:
Test Title: System prevents registration with an already registered phone number	
Description: Test that the system rejects registration if the phone number already exists in the database	

Precondition (If any): Phone number 013******71 must already exist in the database

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the website 2. Click Login Button 3. Click Registration Button 4. Enter Already Registered Phone Number/ Email 5. Enter Password 6. Click Create Account Button	Phone Number: 013******71(Already Registered) Email: user@gmail.com (Already Registered) Password: Newuser#548	Error message is shown: This phone number/Email is already registered. Please log in	As expected,	

Post Condition: No new account is created. Existing account data remains unchanged in the Database.

Test Case 7: Login with Invalid Credentials

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_7</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Login Validation</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: System shows error when login is attempted with wrong password</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test that an appropriate error is shown when incorrect credentials are used to log in</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must have an existing registered account</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Go to the website
2. Enter Registered Phone Number/Email
3. Enter Incorrect Password
4. Click Sign In Button</td><td>Phone Number: 013******71
Email: user@gmail.com
Password: wrongPassword@417</td><td>Error message is displayed: Invalid credentials. Please check your email/phone and password.</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Login attempt is logged. User remains on the Login page.</td></tr>
</table>

Test Case 8: Invalid OTP Entry During Emergency Login

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_8			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Emergency Login Validation			Test Execution date:	
Test Title: System shows error message when wrong OTP is entered				
Description: Test that entering an incorrect OTP during emergency login shows a proper error				
Precondition (If any): User must have requested an OTP for emergency login				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the Emergency Login Page 2. Enter Valid Phone Number 3. Enter Wrong OTP 4. Click Login Button	Phone Number: 013******71 OTP: 000000 (wrong OTP)	Error message is displayed: Invalid OTP. Please try again. User is not logged in	As expected,	
Post Condition: Failed login attempt is logged. User remains on the OTP verification page.				

Test Case 9: Emergency Ambulance Request

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_9</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Emergency Ambulance Service</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: User can request an Ambulance after Emergency Login</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test the Emergency Ambulance Request feature on the Home Page</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must be logged in via Emergency Login</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Login via Emergency Login
2. Navigate to Home Page
3. Click Ambulance Call Button
4. Confirm Current Location
5. Click Request Ambulance Button</td><td>Location: Dhaka, Bangladesh
Phone Number: 013******71</td><td>User should see Ambulance request confirmation with estimated arrival time</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Ambulance request is stored in the database and nearest ambulance is notified.</td></tr>
</table>

Test Case 10: Real-time Ambulance Location Tracking

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_10</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Track Ambulance Location</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: User can track ambulance location in real-time after requesting</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test real-time ambulance location tracking feature after an emergency ambulance request</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must have an active ambulance request</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Login via Emergency Login
2. Request an Ambulance service
3. View tracking screen
4. Observe Ambulance location update</td><td>Request ID: AMB_REQ_001
Location: Dhaka Bangladesh</td><td>Ambulance location is updated on the map every 5 seconds with accurate GPS coordinates</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Real-time location data is continuously fetched from an ambulance GPS device.</td></tr>
</table>

Test Case 11: Nearby Hospital Search

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_11</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Search Nearby Hospitals</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: User can search for nearby Hospitals</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test the Nearby Hospital Search feature from the Home Page</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User must be logged in, and GPS location must be Enabled</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Login to the Application
2. Click on Nearby Hospitals Button
3. Allow Location Access
4. View list of Nearby Hospitals</td><td>Location: Dhaka, Bangladesh</td><td>System displays a list of Nearby Hospitals with distance and availability status</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Search results are displayed based on User's GPS location.</td></tr>
</table>

Test Case 12: Doctor Consultation Booking

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_12			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Telemedicine / Doctor Consult			Test Execution date:	
Test Title: User can book a video consultation with a doctor				
Description: Test the Doctor Consultation Booking feature				
Precondition (If any): User must be registered and logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Click Doctor Consult on Home page 3. Search for a Doctor by specialty 4. Select an available Doctor 5. Choose Date and Time slot 6. Click Confirm Booking Button	Doctor: Cardiologist Date: 15/05/2026 Time: 11.00 am	Booking confirmation is shown and appointment appears in Upcoming Appointments	As expected,	

Post Condition: Appointment details are saved in the database, and reminders are scheduled.

Test Case 13: View Digital Medical Records

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_13			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Digital Medical Records			Test Execution date:	
Test Title: Premium user can view digital medical records				
Description: Test the Digital Medical Records viewing feature for premium users				
Precondition (If any): User must be a premium subscriber and logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Click Medical Records on Home Page 3. View list of stored Medical Records 4. Click on a specific record to open it	User Type: Premium Record: Blood Test Report 18/052026	Medical record opens and displays all health history details within 3 seconds	As expected,	

Post Condition: Record view activity is logged in the system.

Test Case 14: Download Medical Records as PDF

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_14			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Download Digital Medical Records			Test Execution date:	
Test Title: Premium user can download medical record as PDF				
Description: Test the PDF download functionality of digital medical records				
Precondition (If any): User must be a premium subscriber with at least one saved medical record				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Click Medical Records 3. Select Record 4. Click Download as PDF Button	User Type: Premium Record: Prescription 15/05/2026	PDF file is downloaded successfully to the user's device	As expected,	

Post Condition: Download is logged, and PDF is generated from stored medical record data.

Test Case 15: Appointment Reminder Notification

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_15			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Notifications and Alerts			Test Execution date:	
Test Title: User receives Appointment reminder notification				
Description: Test Appointment reminder notifications are sent to patients				
Precondition (If any): User must have an upcoming appointment booked				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Book a Doctor Appointment 2. Wait for the Scheduled Reminder Time (1 Hour Before) 3. Check Mobile/ App notification	Appointment: Dr. Nusrat Jahan Date: 18/05/2026 Time: 10.30 am	Notification is received 1 hour before the appointment with correct details	As expected,	

Post Condition: Notification is marked as sent in the notification logs.				

Test Case 16: Doctor Views Patient Records During Consultation

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_16			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Doctor views Telemedicine / Medical Records			Test Execution date:	
Test Title: Doctor can access patient medical history during a video consultation				
Description: Test that a doctor can view a patient's medical records during an active telemedicine session				
Precondition (If any): A consultation session must be active between doctor and patient				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Doctor logs in with doctor credential 2. Start active Telemedicine Session with patient 3. Click view Patient Records Button	Doctor ID: DOC_001 Patient: Rafiq Ahmed Session ID: SESS_P-1001	Doctor can view the patient's complete medical history during the consultation	As expected,	

Post Condition: Record access is logged with Doctor ID, patient ID, and timestamp				

Test Case 17: Medicine Reminder Notification

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_17	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Medicine reminder notification	Test Execution date:
Test Title: Subscribed user receives medicine reminder notification	
Description: Test that medicine reminder notifications are sent to subscribed users at correct times	
Precondition (If any): User must be subscribed to medicine reminder service and have reminders configured	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Navigate to Medicine Reminders 3. Add Medicine Reminder	Medicine: CardoPlus 50mg Time: 8.00 am Frequency: Daily	Notification is received at the scheduled time with medicine name and dosage	As expected,	

4. Wait for Reminder Notification				
Post Condition: Reminder is marked as sent, and the next reminder is rescheduled.				

Test Case 18: Home Visit Request for Senior Citizens

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_18	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Mobile Healthcare for Seniors	Test Execution date:
Test Title: Senior citizen can request a home doctor visit	
Description: Test the Home Doctor Visit Request feature for senior citizens	
Precondition (If any): User must be registered as a senior citizen patient	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Navigate to Senior Care / Home Visit 3. Enter Home Address	Address: Mirpur 12 Road 5 Dhaka Date: 18/05/2026 Time: 11.00 am	Home visit request is submitted, and healthcare staff receives a real-time alert	As expected,	

4. Select preferred Date and Time 5. Click Request Home Visit Button				
Post Condition: Home visit request is saved in the database and assigned healthcare staff is notified.				

Test Case 19: Search for Doctor by Name

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_19	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Search By Doctor Name	Test Execution date:
Test Title: User can search for a doctor by name from the Home Page	
Description: Test the search functionality for finding a specific doctor by name	
Precondition (If any): User must be registered and logged in	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Click on the Search bar on the Home Page	Search Query: Dr. Nusrat Jahan	Search results display the doctor's name, specialty, hospital, and availability	As expected,	

3. Type Doctor's Name 4. Press Search / Enter Button 5. View Search result				
Post Condition: Search queries and results are logged for analytics purposes.				

Test Case 20: Access Premium Feature Without Payment

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_20	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Access Control	Test Execution date:
Test Title: System blocks access to Digital Medical Records for non-premium users	
Description: Verify that a free/regular user cannot access the Digital Medical Records feature without purchasing a premium plan	
Precondition (If any): User must be registered and logged in with a free (non-premium) account	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application 2. Navigate to the Home Page	User Type: Free Account: user@gmail.com	System displays an upgrade/payment prompt: This is a premium feature.	As expected,	

3. Click on Medical Records Button 4. Observe System Response		Please purchase a plan to access your medical records. Access is denied.		
Post Condition: No medical record data is exposed to the free user. Payment prompts are displayed correctly.				

Test Case 21: Select Premium Plan and Proceed to Payment

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_21	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Plan Selection	Test Execution date:
Test Title: User can select a premium plan and redirected to the payment page	
Description: Test that the user can choose a premium subscription plan and is correctly redirected to the payment gateway selection screen	
Precondition (If any): User must be logged in with a free account and have clicked on a premium feature	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the Application	Plan: Monthly Premium	User is redirected to the payment gateway selection page	As expected,	

2. Click Medical Records (Premium prompt appears) 3. Click Upgrade / View Plans Button 4. Select a Plan (e.g.: Monthly Premium) 5. Click Proceed to Payment	Price: BDT 299 / month	showing available options (Bkash, Nagad, Rocket, Visa, Mastercard)		
Post Condition: Selected plan details (plan name, price, duration) are stored in session for payment processing.				

Test Case 22: Successful Payment via Bkash

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_22	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Bkash	Test Execution date:
Test Title: User successfully completes payment using Bkash	
Description: Test that a user can complete a premium plan purchase using Bkash and the account is upgraded too premium	
Precondition (If any): User must be on the payment gateway selection page with a plan selected; valid Bkash account required	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Select Bkash as Payment Method 2. Enter Bkash Mobile Number 3. Enter Bkash PIN / OTP 4. Confirm Payment 5. Observe result after Payment	Payment Method: Bkash Bkash Number: 013******71 Amount: BDT 299	Payment success screen is shown with transaction ID. User account is upgraded too Premium. Medical Records feature becomes accessible.	As expected,	
Post Condition: Transaction records are stored in the database with the status of 'SUCCESS'. User roles are updated too Premium.				

Test Case 23: Successful Payment via Nagad

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_23	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Nagad	Test Execution date:
Test Title: User successfully completes payment using Nagad	
Description: Test that a user can complete a premium plan purchase using Nagad and the account is upgraded too premium	
Precondition (If any): User must be on the payment gateway selection page with a plan selected; valid Nagad account required	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select Nagad as Payment Method 2. Enter Nagad Mobile Number 3. Enter Nagad PIN / OTP 4. Confirm Payment 5. Observe result after Payment	Payment Method: Nagad Nagad Number: 013******71 Amount: BDT 299	Payment success screen is shown with transaction ID. User account is upgraded too Premium. Medical Records feature becomes accessible.	As expected,	
Post Condition: Transaction records are stored in the database with the status of 'SUCCESS'. User roles are updated too Premium.				

Test Case 24: Successful Payment via Visa/Mastercard

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_24	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Card	Test Execution date:
Test Title: User successfully completes payment using Visa or MasterCard	
Description: Test that a user can complete premium plan purchase using a debit/credit card and account is upgraded	

Precondition (If any): User must be on the payment gateway selection page with a valid Visa/Mastercard				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select Visa / MasterCard as Payment Method 2. Enter card Number 3. Enter expiry Date and CVV 4. Enter card Holder Name 5. Click Pay now 6. Complete OTP verification (if required)	Card Number: 4*** **** **** 1234 Expiry: 12/27 CVV: *** Amount: BDT 299	Payment success screen is shown with transaction ID. User account is upgraded too Premium. Medical Records feature becomes accessible.	As expected,	
Post Condition: Transaction records are stored in the database with the status of 'SUCCESS'. User roles are updated too Premium.				

Test Case 25: Failed Payment – Insufficient Balance

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_25	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Heigh	Test Executed by:
Module Name: Payment / Failure Handling	Test Execution date:
Test Title: System shows appropriate error when payment fails due to insufficient balance	

Description: Test that the system properly handles a payment failure caused by insufficient balance in the user's payment account				
Precondition (If any): User must be on the payment page with a payment account that has insufficient balance				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select Payment Method (e.g.: Bkash) 2. Enter valid Phone Number with Low Balance 3. Enter Correct PIN / OTP 4. Click Confirm Payment 5. Observe System Response	Payment Method: Bkash Bkash Number: 013******71 Account Balance: BDT 50 (less than BDT 299)	Payment failure message is shown: Payment failed due to insufficient balance. Please try again. User account remains free.	As expected,	
Post Condition: Transaction is logged with status 'FAILED'. No premium access is granted. Users can retry with a different method.				

Test Case 26: Payment Cancelled by User

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_26	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Payment Cancellation	Test Execution date:

Test Title: User can cancel payment and return to the plan selection page				
Description: Test that a user who cancels during the payment process is safely returned to the application without losing account data				
Precondition (If any): User must be on the payment gateway page with a plan selected				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select a plan and proceed to Payment 2. Select Payment Method 3. Click Cancel / Back Button before confirming Payment 4. Observe System Response	Plane: Monthly Premium Payment Method: Nagad (Not Completed)	User is returned to the plan selection page. No charge is made. Account remains free. A message appears: Payment cancelled. No charges were made.	As expected,	
Post Condition: Transaction is logged with the status 'CANCELLED'. User account type remains unchanged.				

Test Case 27: Payment Receipt / Invoice Generation

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_27	Test Designed date: 10/05/2026

Test Priority (Low, Medium, High): Medium	Test Executed by:
Module Name: Payment Receipt	Test Execution date:
Test Title: User receives a payment receipt after successful premium purchase	
Description: Test that a payment receipt or invoice is generated and accessible after a successful premium subscription payment	
Precondition (If any): User must have successfully completed a premium plan payment	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Complete a successful Payment 2. Navigate to Profile or Payment History 3. Click View Receipt / Invoice 4. Observe Receipt details	Transaction ID: TXN_2026_050 Plan: Monthly Premium Amount: BDT 299 Date: 18/05/2026	Receipt is displayed with Transaction ID, Plan name, Amount paid, Payment method, Date, and Username. Option to download receipt as PDF is available.	As expected,	

Post Condition: Receipt data is permanently stored in the database and linked to the user's account.

Test Case 28: Language Input Field Selector

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: FR_28	Test Designed date: 10/05/2026

<table>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Language Input Field Selector</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: Verify that clicking 'Select Language' opens the language picker modal</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test that the language selector button in the top navbar opens the bottom-sheet modal with all available languages</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User has opened the MediCare landing page (landing.html) in a browser</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Open landing.htm
2. Locate the 'Select Language' button in the top navbar
3. Click the 'Select Language' button</td><td>No input required</td><td>A bottom-sheet modal slides up showing 5 language options: English, Bangla, Hindi, Arabic, French.</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Language picker modal is visible. No language change has been made yet</td></tr>
</table>

Test Case 29: Language Selection

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_29</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): Medium</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Language Options</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: User can change interface language to Bengali</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test the Language Selection feature in user settings</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any):</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Click Language Option
2. Select Bengali (Bangla)
3. Click Save Changes</td><td>Language Selected: Bengali</td><td>Application interface switches to Bengali language and preference is saved</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: Language preference is stored in the user's profile in the database.</td></tr>
</table>

Test Case 30: Landing Page – Get Started Button

<table>
<tr><td colspan="3">Project Name: MediCare</td><td colspan="2">Test Designed by: MD. Mridul Ali</td></tr>
<tr><td colspan="3">Test Case ID: FR_30</td><td colspan="2">Test Designed date: 10/05/2026</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): Medium</td><td colspan="2">Test Executed by:</td></tr>
<tr><td colspan="3">Module Name: Landing Page – Get Started</td><td colspan="2">Test Execution date:</td></tr>
<tr><td colspan="3">Test Title: Verify that clicking 'Get Started' navigates to the phone number screen</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test that the primary CTA button in the hero section routes the user to screen1-phone.html</td><td colspan="2"></td></tr>
<tr><td colspan="5">Precondition (If any): User is on the landing page. All HTML files are in the same directory.</td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Open landing.html
2. Locate the 'Get Started' button in the hero section
3. Click the 'Get Started' button</td><td>No input required</td><td>Browser navigates to Emergency Login Page (Phone number entry page).</td><td>As expected,</td><td></td></tr>
<tr><td colspan="5">Post Condition: User is redirected to the phone number entry screen to begin the login/OTP flow.</td></tr>
</table>

Test Case 31: Landing Page – Learn More Button

Project Name: MediCare			Test Designed by: MD. Mridul Ali	
Test Case ID: FR_31			Test Designed date: 10/05/2026	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Landing Page – Learn More Button			Test Execution date:	
Test Title: Verify that clicking 'Learn More' smoothly scrolls to the features section				
Description: Test that the secondary CTA button triggers a smooth scroll to the 'Why patients trust us' section.				
Precondition (If any): User is on the landing page. The features section exists below the fold.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open landing.html 2. Locate the 'Learn More' button in the hero section 3. Click the 'Learn More' button	No input required	Page smoothly scrolls down to the 'Why Patient Trust Us' features section.	As expected,	
Post Condition: The features section is now visible in the viewport				

Non-Functional Test Case:(QA Suggested)

Test Case 32: Security Check

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: NFR_32	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Security – Access Control	Test Execution date:
Test Title: Unauthenticated or wrong-role users cannot access another patient's digital medical records.	
Description: Verify the system blocks any attempt to view private medical records without a valid authenticated session, and that Patient B cannot access Patient A's data even while logged in.	
Precondition (If any): Two registered accounts exist (Patient A & Patient B). Patient A has at least one medical record stored. The staging server enforces HTTPS.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Log in as Patient A, open a medical record, and copy its URL from the browser address bar	Patient A URL: /records/patient-A-001	URL is noted for re-use in later steps	As expected	
2. Log out of Patient A's session completely and verify the session cookie is cleared	N/A	User is redirected to the Login page; session cookie is invalidated	As expected	
3. Without logging in, paste Patient A's record	URL: /records/ patient-A-001 (unauthenticated)	System returns HTTP 401 Unauthorized or redirects to Login page;	As expected	

URL directly into the browser and press Enter		zero medical data is displayed		
4. Log in as Patient B and manually navigate to Patient A's record URL	Patient B: user2@gmail.com @789	System returns HTTP 403 Forbidden; Patient A's records are NOT visible to Patient B	As expected	
5. Attempt an IDOR attack by incrementing the patient ID in the URL (e.g., patient-A-002, patient-A-003)	Modified URLs: /records/patient-A-002 /records/patient-A-003	All attempts are denied; no data from any other patient is exposed; OWASP ZAP reports 0 critical issues	As expected	
Post Condition: All attempts are denied; no data from any other patient is exposed; OWASP ZAP reports 0 critical issues				

Test Case 33: Performance Check

Project Name: MediCare	Test Designed by: MD. Mridul Ali
Test Case ID: NFR_33	Test Designed date: 10/05/2026
Test Priority (Low, Medium, High): High	Test Executed by:
Module Name: Performance – Emergency Services	Test Execution date:
Test Title: Emergency service dashboard must fully load within 2 seconds; ambulance GPS must refresh every 5 seconds	
Description: Measure the page load time of the emergency dashboard immediately after emergency login and verify that real-time ambulance location updates arrive every 5 seconds, under standard network conditions.	

Precondition (If any): Apache JMeter is installed and configured. A valid test phone number exists. The staging environment is running. GPS simulation is active.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Configure JMeter HTTP Request Sampler targeting the emergency login endpoint; authenticate and retrieve a session token	Endpoint: POST /api/emergency-login Phone: 013******71 OTP:589741	Session token received; JMeter records Time to First Byte (TTFB)	As expected	
2. Using the session token, send a GET request to the emergency dashboard endpoint and record total page load time	Endpoint: GET /emergency-dashboard Session: valid token	Full page (ambulance button, hospital list, map widget) loads in ≤ 2,000 ms	As expected	
3. Repeat the load request 10 times and calculate average, minimum, and maximum response times	Iterations: 10 Sampler: HTTP Loop Count: 10	Average ≤ 2,000 ms; maximum ≤ 3,000 ms; no timeouts	As expected	
4. After requesting an	Request ID: AMB_REQ_001 Location: Dhaka	Ambulance location pin refreshes on the map every	As expected	

ambulance, observe the GPS tracking screen and measure the time between consecutive location pin updates on the map		5 seconds (±1 second tolerance)		
5. Run the combined load and GPS refresh test for 5 continuous minutes and check for degradation	Duration: 5 min Users: 10 concurrent	Load time stays ≤ 2,000 ms throughout; GPS refresh interval stays ≤ 6 seconds; zero crashes	As expected	
Post Condition: JMeter Response Time report and GPS refresh log are saved. Any result exceeding thresholds is filed as a Performance defect in Jira with severity High.				

8. ITEM PASS/FAIL CRITERIA

The MediCare system testing process will follow specific pass/fail criteria to determine whether each test item satisfies the expected requirements. A test item will be marked as "Pass" only if the actual result matches the expected result without any critical issues. Otherwise, the test item will be marked as "Fail".

Criteria Type	**Pass Criteria**	**Fail Criteria**
Functional Testing	All functions work according to system requirements	Functions produce incorrect or unexpected output
Login & Authentication	Valid users can log in securely and invalid users are rejected	Unauthorized access is possible or valid login fails

Emergency Services	Ambulance and hospital services respond within required time	Delayed response or request failure
Performance Testing	System handles at least 1000 concurrent users without crash	System slowdown, crash, or timeout occurs
Security Testing	No critical vulnerabilities found; encryption works properly	Data leakage, unauthorized access, or security flaws found
Database Testing	Data is stored, updated, and retrieved correctly	Data corruption or incorrect database operations occur
Usability Testing	Users can complete tasks easily without confusion	Navigation problems or unclear interface detected
Compatibility Testing	System works properly on supported browsers/devices	UI breaks or features fail on supported devices
Acceptance Testing	Client confirms requirements are satisfied	Client rejects functionality or requirements are incomplete

Table 5: Item Pass / Fail Criteria

Test Completion Criteria:

The testing phase will be considered successfully completed when:

- At least 95% of all test cases pass successfully.
- No critical or high-severity defects remain unresolved.
- All major functional requirements are verified.
- Security and performance requirements meet expected standards.
- Client acceptance testing is approved.

9. TEST DELIVERABLES

The following documents and artefacts will be produced and formally delivered as part of the MediCare software testing process. All deliverables must be reviewed and signed off by the QA Team Lead before release.

Deliverable	Description
Software Test Plan	Complete testing strategy, scope, schedule, and testing process

Test Cases/Test Items	Detailed functional and non-functional test cases
Test Data	Input values and datasets used for testing
Bug/Defect Reports	Reports containing detected issues and bug severity
Test Execution Reports	Results of executed test cases
Performance Test Report	Load and performance testing outcomes
Security Test Report	Vulnerability and security assessment results
Meeting Minutes (MoM)	Records of project and testing meetings
Requirement Traceability Matrix (RTM)	Mapping between requirements and test cases
Final Test Summary Report	Overall testing summary and project quality status
User Acceptance Test (UAT) Report	End-user/client acceptance results

Table 6: Test Deliverables

10. STAFFING AND TRAINING NEEDS

The MediCare project requires skilled team members and proper training to ensure effective development and testing activities.

10.1 Staffing Requirements

Role	**Required Number**	**Responsibility**
Project Manager	1	Project planning and management
Frontend Developer	3	User interface development
Backend Developer	3	API and server-side development
UI/UX Designer	1	Design application interface

Database Administrator	1	Database management and security
QA/Test Engineer	3	Software testing and bug reporting
DevOps/Cloud Engineer	1	Deployment and server maintenance
Technical Support Staff	1	User and system support

Table 7: Staffing Requirements

10.2 Training Needs

The project team members may require the following training:

Training Area	Purpose
Playwright Automation Testing	Automated functional testing
Apache JMeter	Performance and load testing
OWASP ZAP	Security vulnerability testing
API Testing with Postman	Backend API validation
Git & GitHub Workflow	Version control and collaboration
Cloud Deployment	Server deployment and maintenance
Healthcare Data Privacy	Understanding HIPAA/GDPR compliance
Emergency Response Workflow	Handling emergency medical service logic

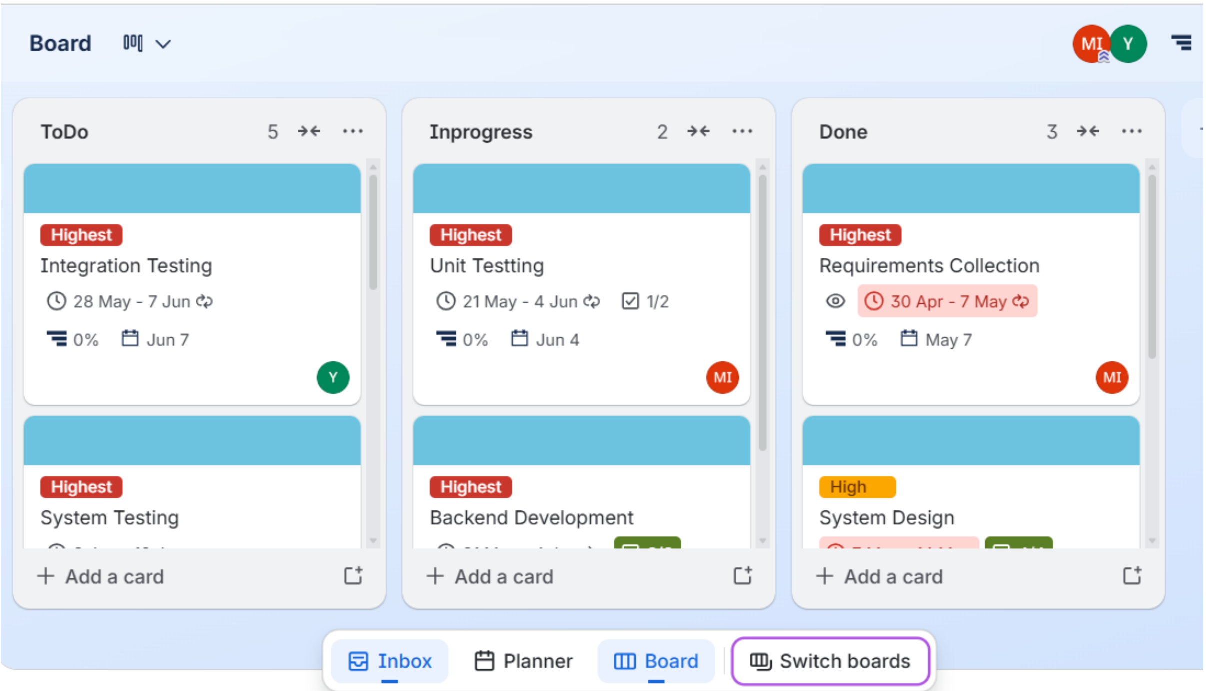
Table 8: Training Needs

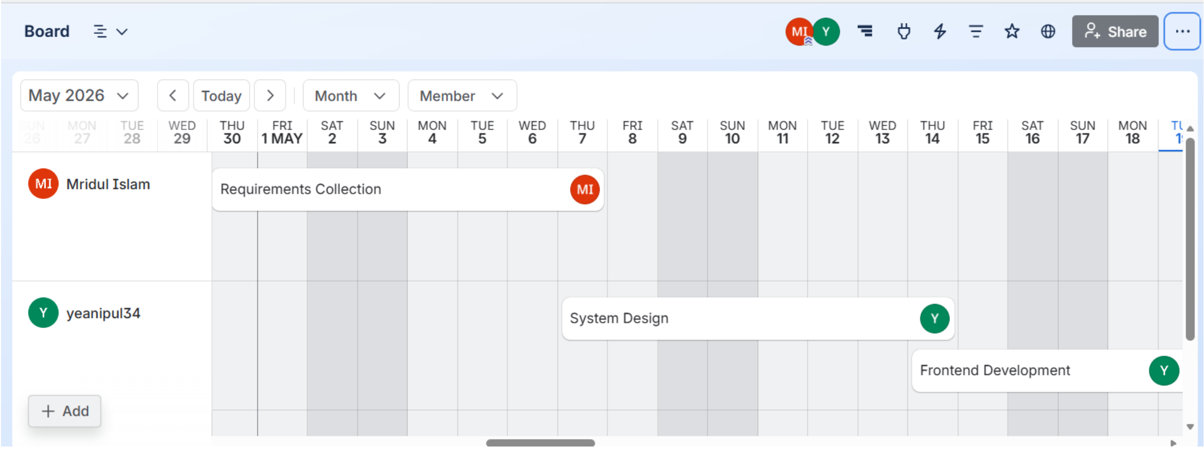
11. RESPONSIBILITIES

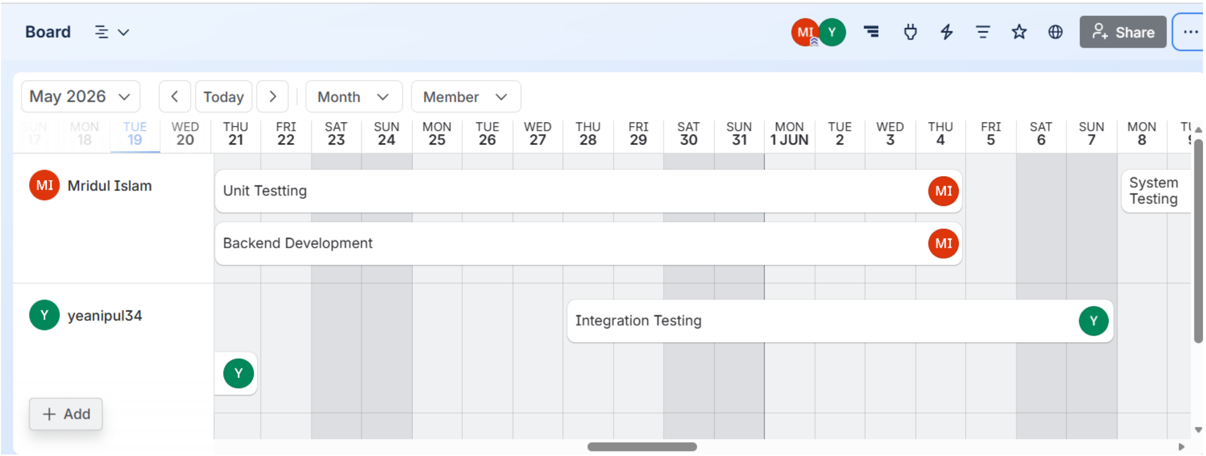
	TM	PM	Dev Team	Test Team	Client
Acceptance test Documentation & Execution	X	X		X	X
System/Integration test Documentation & Exec.	X		X	X	
Unit test documentation & execution	X		X	X	
System Design Reviews	X	X	X	X	X
Detail Design Reviews	X	X	X	X	
Test procedures and rules	X	X	X	X	
Screen & Report prototype reviews			X	X	X
Change Control and regression testing	X	X	X	X	X

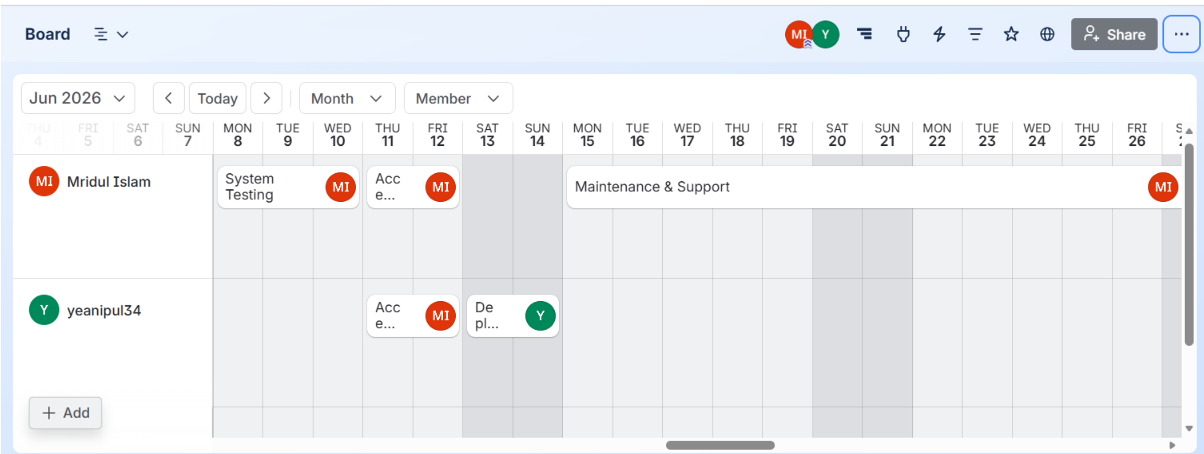
Table 9: Responsibilities

12. TESTING SCHEDULE







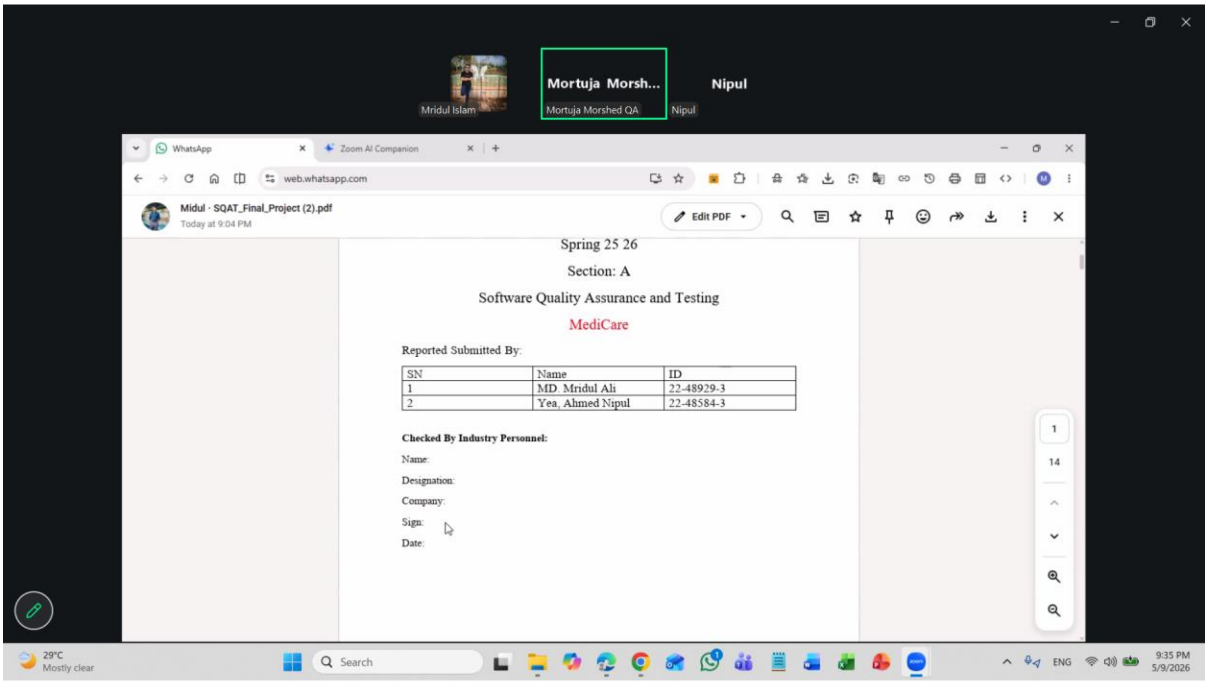


13. PLANNING RISKS AND CONTINGENCIES

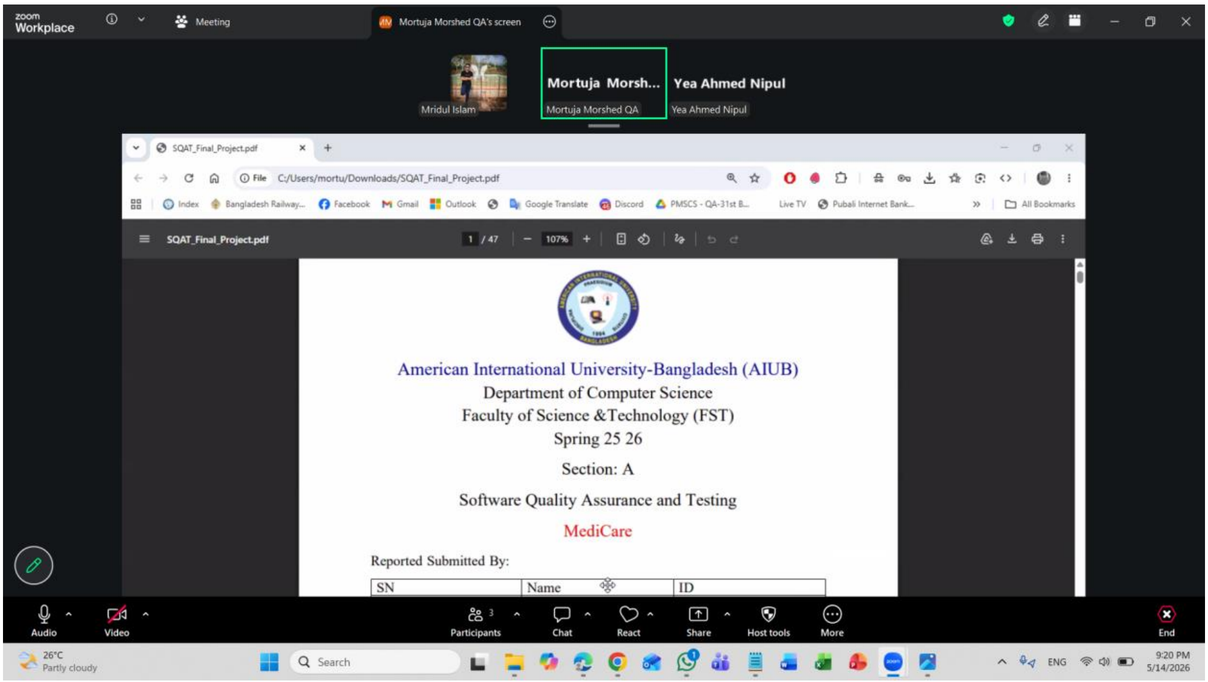
The MediCare project may face several technical, operational, and management risks. Proper contingency plans are prepared to minimize project impact.

Risk	Impact	Probability	Contingency Plan
Server Failure	System downtime	Medium	Use backup server and automatic recovery system
Security Breach	Sensitive data exposure	Medium	Apply SSL/TLS encryption and regular security testing
Internet Connectivity Issues	Service interruption	High	Use retries mechanisms and offline data caching
Payment Gateway Failure	Payment processing interruption	Medium	Provide multiple payment gateway options
GPS/Location Service Failure	Ambulance tracking issues	Medium	Enable manual location sharing
High User Traffic	System slowdown or crash	High	Use cloud auto-scaling and load balancing
Third-Party API Failure	External services unavailable	Medium	Maintain fallback APIs and retry policies
Delayed Development	Project timeline extension	Medium	Add buffer time and weekly progress monitoring
Data Loss	Loss of patient information	Low	Daily automatic database backup
Team Member Unavailability	Development/testing delays	Medium	Cross-training and task redistribution
Critical Software Bugs	System instability	Medium	Immediate bug fixing and emergency patch release
Miscommunication Between Teams	Requirement misunderstanding	Medium	Conduct regular meetings and maintain documentation

Online Meeting with QA:



Date: 09/05/2026



Date: 14/05/2026

---------------The End--------------